

4 Fair Division

Definition 4.1 (Fair Division Method) A fair division method is a procedure that can be followed that will result in a division of the items in a way so that each party feels they have received their fair share. We make the following assumptions:

- The parties are non-cooperative, so the method must operate without communication between parties.
- The parties have no knowledge of what the other players like
- The parties act rationally, and do not make emotional decisions.
- The method should allow the parties to make a fair division without an outside arbitrator.

Definition 4.2 (Fair Share) When N parties divide something equally, each party's fair share is the amount they are entitled to.

Example 4.3 Suppose that 4 classmates are splitting equally a \$12 pizza that is half pepperoni, half veggie that someone else bought them. What is each person's fair share?

Fair share is $1/N$.

Note that the dollar amount for each player is \$3, but each party might value portions of the pizza differently, for ex: vegetarian...

Example 4.4 Suppose that in the above example, Steve likes pepperoni twice as much as veggie. Describe a fair share for Steve.

Note that there are many different answers which are correct here.

4.1 Divider Chooser Method

Definition 4.1.1 (Divider-Chooser Method)

1. The divider cuts the item into two pieces that are, **in his eyes**, equal in value.
2. The chooser selects either of the two pieces.
3. the divider receives the remaining piece.

Example 4.1.2 Troy and Abed are sharing a \$12 Subway sandwich, which is a foot long and half veggie and half chicken. Troy likes chicken 2 times as much as veggies, and Abed likes chicken and veggies equally. If Troy is the divider, describe how he will split the sandwich (only vertical slices are allowed). Which part will Abed choose? Describe the perceived value of each piece to its owner.

Troy makes a cut 4.5" into the chicken side.
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Example 4.1.3 Your turn! Annie and Jeff are now going to share a sandwich, which is half veggie and half chicken, and is a foot long. The sandwich costs \$16. Annie is a vegetarian, while Jeff likes veggie 3 times as much as he likes chicken. If Jeff is the divider, describe how he will split the sandwich (only vertical slices are allowed). Which part will Annie choose? Describe the perceived value of each piece to it's owner.

Jeff cuts $\frac{2}{3}$ (4") into the veggie side.
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